Cluster 1



Video 1, 01:40



Video 5, 05:50



Video 103, 06:40

Cluster 2



Video 10, 05:32



Video 18, 05:38

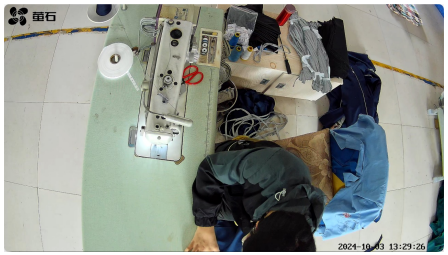


Video 106, 03:58

Cluster 3



Video 18, 01:53



Video 94, 01:54



Video 95, 02:10

Workplace-video examples of the three motion clusters in zipper sewing.

Rows show selected frames from video segments associated with each cluster. Cluster 1 primarily involved repeated material propulsion. Cluster 2 involved material retrieval, removal, and replacement. Cluster 3 included brief pauses, adjustments, off-task behavior, and temporary absence and was not retained as a separate operational phase. Cluster 1 primarily corresponded to P2, whereas Cluster 2 contributed to both preparation (P1) and removal and return (P3). Video identifiers and timestamps (min:s) appear below each frame.